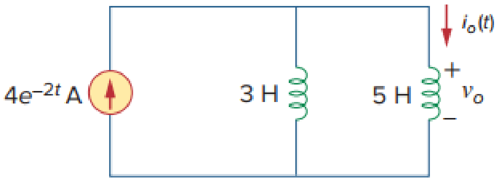
Chapter 6, Problem 60P

Problem

In the circuit of Fig. 6.82, *io*(0) = 2 A. Determine *io*(*t*) and *vo*(*t*) for *t* > 0.

Fig. 6.82:



Step-by-step solution

Step 1 of 2

Refer to the circuit diagram depicted in Figure 6.82 in the textbook.

The value of source current is, $i_s(t) = 4e^{-2t}$ V

In Figure 6.82, two indictors are connected in parallel. Find its equivalent inductance.

$$\begin{aligned} L_{eq} &= 3 \| 5 \\ &= \frac{3\times 5}{3+5} \\ &= \frac{15}{8} \text{ H} \end{aligned}$$

Determine the voltage, $v_o(t)$.

$$\begin{aligned} v_o(t) &= L_{eq}\frac{di_s(t)}{dt} \\ &= \frac{15}{8}\frac{d}{dt}\left(4e^{-2t}\right) \\ &= \frac{15}{8}\left(-4\times 2e^{-2t}\right) \\ &= -15e^{-2t} \text{ V} \end{aligned}$$

Therefore, the voltage $v_o(t)$ is $\boxed{-15e^{-2t}\text{ V}}$.

Step 2 of 2

Determine the current, $i_o(t)$ through 5 H inductor.

$$i_o(t) = \frac{1}{L}\int_0^t v_o(t)\,dt + i_o(0)$$

$$= \frac{1}{5}\int_0^t -15e^{-2t}\,dt + 2$$

$$= \frac{-15}{5}\int_0^t e^{-2t}\,dt + 2$$

$$= -3\left(\frac{e^{-2t}}{-2}\bigg|_0^t\right) + 2$$

$$= 1.5\left(e^{-2t} - 1\right) + 2$$

$$= 1.5e^{-2t} - 1.5 + 2$$

$$= 1.5e^{-2t} + 0.5\text{ A}$$

Therefore, the current $i_o(t)$ is $\boxed{1.5e^{-2t} + 0.5\text{ A}}$.